INSTRACTA ELEARNING SOLUTIONS

INSTRACTA RESEARCH PAPERS, NO. 3 · PILLAR: LOCAL

Built Where It Is Used

Language, pedagogy, open technology and local partnership as conditions of effective digital learning in Africa

Opiyo Michael Ouru
Instracta eLearning Solutions, Nairobi, Kenya
October 2026 · Working paper. Not peer reviewed.

ABSTRACT

Education reforms and technologies imported from elsewhere have a long record of looking successful while failing to change what happens in classrooms. This paper asks what it means for digital learning to be local, and whether locality is a matter of sentiment or of effectiveness. It distinguishes three layers of localisation: language and content, pedagogy, and institutions, including ownership, capability and data. For each layer it reviews evidence from Kenya, Sub-Saharan Africa and the wider development literature. It finds that language mismatch is a major and well-documented barrier to learning, that Kenyan trials combining books in learners' languages with in-classroom teacher support improved reading, that pedagogical models promoted by international agencies have often been adopted in form without being sustained in practice, and that the Kenyan and continental policy frameworks now give providers a clear mandate to build locally. The paper argues that open-source technology, open licensing and local partnerships are not ideological preferences but the practical means by which institutions gain the ability to adapt, host and sustain what they use. It proposes five tests of whether a digital learning programme is local in substance, notes the risks of the position, and states how Instracta's products and partnerships are intended to meet the tests.

Keywords: localisation; language of instruction; pedagogy; open source; digital public goods; Kenya; Africa

1. Introduction

Development practice has a recurring pattern. A model that works in one place is exported to another, its visible features are adopted, and its results do not follow. Pritchett and Woolcock (2004) described how, in the 1960s and 1970s, developing countries copied developed-country models built on standardised, top-down programmes run by centralised civil services, an approach the authors judged to have failed in nearly every sector. Education technology is a recent instance. Platforms, curricula and pedagogies are carried across borders because they are available, well funded and well marketed, and then found not to fit the language, infrastructure, teachers or institutions that receive them.

The response is often to say that technology should be local. The word is rarely defined, and it is easily taken for a matter of branding, of putting a flag or a local name on an imported product. This paper takes the claim more seriously and asks whether locality has a measurable bearing on whether learners learn. Section 2 distinguishes three layers of localisation. Sections 3 to 5 review the evidence on each. Section 6 turns to policy frameworks and open technology. Section 7 proposes five tests. Section 8 discusses risks and limits. The author is the founder of a Kenyan provider with an interest in the conclusion, and the argument should be read with that in mind.

2. Three layers of localisation

We distinguish three layers. They differ in how easily they are seen and how hard they are to change.

1. Language and content. Whether the learner can understand the medium of instruction, and whether examples, cases and assessments reflect the learner's world.
2. Pedagogy. Whether the way of teaching suits the classroom, the teacher's training and the learner's expectations, or has been adopted because a funder or reform agenda required it.
3. Institutions. Who owns the platform and the content, who can host, adapt and repair it, where the data are held, and who is accountable to learners.

Most programmes that claim to be local address the first layer, partially. The argument of this paper is that effectiveness and sustainability depend more on the second and third.

3. Language and content

The evidence that language of instruction affects learning is among the strongest in comparative education. The 2016 UNESCO policy paper, If you don't understand, how can you learn?, estimated that 40% of the world's population does not have access to education in a

language they speak or understand, and illustrated the consequence with data from Côte d'Ivoire, where 55% of grade 5 students who spoke the language of the test at home reached basic reading in 2008, against 25% of those who did not (UNESCO, 2016). Ouane and Glanz (2011), in an analytical review for the UNESCO Institute for Lifelong Learning and the Association for the Development of Education in Africa, argued that language choices should be addressed across policy, costing, governance, classroom practice, literacy and publishing, including the role of locally based multilingual publishing.

Kenya's constitution establishes Kiswahili as the national language, Kiswahili and English as the official languages, and an obligation on the state to promote and protect the diversity of the languages of the people (Republic of Kenya, 2010, Art. 7). The Basic Education Act of 2013 and the competency-based curriculum operate within that framework (Republic of Kenya, 2013). For a provider, the practical question is which languages to build in. Kenyan evidence is encouraging on the value of materials in the languages learners use. The randomised evaluation of the PRIMR Initiative, in more than 400 schools in three counties, supplied books in English and Kiswahili alongside ongoing instructional support and classroom observation, and found improved oral reading fluency in both languages (Piper et al., 2014). The finding does not show that any one language is best. It shows that a package built around the languages in use, with support in the classroom, can move reading outcomes at scale.

Content is the less-studied half of this layer. Ladson-Billings (1995) argued that culturally relevant pedagogy requires that students experience academic success, develop cultural competence, and develop a critical consciousness about the social order. Her work concerned African American students in the United States, and its transfer is by analogy, but the principle that examples and cases should connect to learners' lives is consistent with the evidence on teaching to the learner's level discussed in the second paper in this series. Ngũgĩ wa Thiong'o (1986) made the broader argument that a language of instruction carries a world-view, and that learning in a colonial language can separate education from the community it is meant to serve. For a digital provider, the practical consequences are modest and concrete: use local case material, work with subject experts and language editors who live in the context, and test comprehension with the learners for whom the material is intended.

4. Pedagogy: the limits of importing a way of teaching

Learner-centred education is the clearest example of a pedagogy promoted internationally with limited regard for fit. Schweisfurth (2013), in a book-length review, asked whether the approach suits every society and classroom, noting that supporters tie it to cognitive psychology and to the aims of democratic citizenship and a skilled workforce, while critics question whether it fits every cultural and resource context and point to persistent implementation problems in the Global South. Tabulawa (2003) argued that the appeal of learner-centred pedagogy to international aid agencies was mainly political and ideological, and that it has been promoted as a worldview intended to develop a preferred kind of society,

although it is usually justified in cognitive and educational terms. These are contested claims, and the authors do not argue that learner-centred methods lack value. Their point is that adoption by decree, without the conditions that make a method workable, produces form without substance.

Pritchett, Woolcock and Andrews (2013) gave the pattern a name, isomorphic mimicry: the adoption of the forms of functional institutions without their functions, so that a system comes to look like a successful one while performing no better. The later book by Andrews, Pritchett and Woolcock (2017) developed the argument that capability is built by locally driven problem-solving and iterative adaptation rather than by transplanting best practice. For an education provider, the lesson is to resist the supposition that a model validated elsewhere is validated here. The evidence on what works, reviewed in the second paper, is broadly consistent across settings on two points, targeting instruction to the learner's level and supporting teachers over time. Beyond those, the safest course is to test locally and adapt.

The teacher is the point where pedagogy meets context. Bold and colleagues (2017), using nationally representative data from seven Sub-Saharan African countries covering close to 40% of the region's population, found that students received about two hours and fifty minutes of teaching per day, just over half of scheduled time, and that teacher knowledge of content and pedagogy was limited and strongly associated with student achievement. Hennessy and colleagues (2022) reviewed technology-mediated teacher professional development in low- and middle-income countries and noted that professional development there is often limited and short-lived. A locally effective programme is therefore one that builds the capacity of local teachers and trainers and does not bypass them. The Kenyan trials discussed above, which paired materials with in-classroom coaching and observation, support the same conclusion (Piper et al., 2014; Piper et al., 2018).

5. Institutions: ownership, capability and data

The third layer is the least visible and the most consequential for sustainability. A programme that depends on a foreign platform, a proprietary licence or an overseas server can be switched off by decisions that local institutions do not control. The Global Education Monitoring Report of 2023 framed the issue as one of terms: whose interests technology serves, who frames the problems it is meant to solve, and whether it is appropriate, equitable, evidence-based and sustainable (UNESCO, 2023). Among its recommendations to governments were independent evaluation of education technology, pilot projects that reflect total cost of ownership, and transparency in spending and contracts with private companies (UNESCO, 2023).

Data are a direct case. The Data Protection Act of 2019 gives individuals in Kenya rights over their personal data and places obligations on data controllers and processors, including the registration of some of them with the Office of the Data Protection Commissioner (Republic of

Kenya, 2019). A learning platform that holds the records of Kenyan learners operates within that regime. Providers that host data in-country, that can export a learner's records, and that collect only what is needed are better placed to comply and to earn trust.

Ownership is addressed by openness. The Digital Public Goods Standard lists as baseline indicators open licensing, clear ownership, platform independence, documentation, a mechanism for extracting data, and adherence to privacy laws and to standards and best practices (Digital Public Goods Alliance, n.d.). The UNESCO Recommendation on Open Educational Resources, adopted by the member states of UNESCO in November 2019, encourages the creation, adaptation and sharing of openly licensed educational materials (UNESCO, 2019). The logic is practical. Open licences and open-source software allow an institution in Kenya to host, translate, adapt and repair what it uses without negotiating with a vendor, and allow capability to accumulate locally. They do not guarantee quality or sustainability, which depend on maintenance and community, but they remove a structural barrier.

6. Policy frameworks that make the case

Local provision is not only a preference of providers. It is where policy points. The African Union's Continental Education Strategy for Africa 2016 to 2025 sets out continental goals for education and training, and its Digital Transformation Strategy for Africa 2020 to 2030 sets a framework for digital development across the continent (African Union, 2016; African Union, 2020). Kenya's National Digital Masterplan 2022 to 2032, launched in April 2022, includes targets to extend fibre to 40,000 public schools and learning institutions, to establish 1,450 digital innovation hubs, and to develop ICT skills and local software and manufacturing capacity, including two software factories (ITWeb Africa, 2022). These documents create demand for locally built and locally hosted digital learning, and a standard against which providers can be asked to show how their work contributes.

The policy framework also sets a trap. Frameworks can be satisfied by statements and not by capability, which is the form-without-function that the development literature warns against. Providers that wish to be credible should therefore state specific, checkable commitments, such as the proportion of staff, partners and content creators based in the country, the languages in which materials are available, and the share of intellectual property held openly.

7. Five tests of whether a programme is local in substance

Drawing on the three layers, we propose five tests that a funder, partner or learner can apply to any digital learning programme. They are intended to be answerable with evidence and not with assurances.

Table 1. Five tests of local substance

Test	Question	Evidence to look for
1. Language	Are materials available in the languages learners use, and have learners shown they understand them?	List of languages; comprehension results; use of local language editors
2. Fit	Was the pedagogy tested with local learners and teachers, and adapted as a result?	Pilot records; documented changes; teacher feedback
3. People	Do local teachers, trainers and subject experts design and deliver, and does the programme build their capacity?	Staff and partner composition; training records; local evaluators
4. Ownership	Can a local institution host, adapt and exit the platform with its content and data intact?	Open licences; export formats; hosting arrangements; data processing agreements
5. Accountability	Is the programme answerable to local learners and institutions, and is its evaluation independent?	Advisory structures; published results; independent evaluator

The tests are not a scoring system. They are a way of asking, of any programme, which layers have been addressed and which are merely asserted. A programme may pass some and fail others. The value lies in naming which.

8. Risks, limits and Instracta's position

The argument for local provision carries risks that should be stated. Local is not the same as good: a locally built product can be poor, and foreign designs sometimes outperform local ones. Insistence on locality can slide into parochialism or into essentialising a culture, treating learners as members of a type and not as individuals. Open licences can be adopted for appearance while the work of maintenance is neglected. And the evidence for the specific claim that locality causes better learning is thin. What the literature supports is narrower: that language mismatch harms learning, that teacher support and instruction matched to the learner work across contexts, that imported reforms often fail to take hold, and that institutional ownership is a precondition for sustainability. The causal contribution of any single local feature is not established.

Instracta's position against the five tests is partial. It is a Kenyan organisation based in Nairobi and publishes its Moodle theme as open source. Its authoring tool produces content that can be exported and hosted elsewhere, a precondition for the fourth test. Its learning management system is in preparation and is planned to be built on Moodle, an open-source platform, and its accessibility plugin is planned. It has not yet published a list of languages in which its materials are available, a data processing statement, or a record of local pilots with documented adaptations, and it should do so. It has proposed an independent evaluator and an advisory board, whose appointment is a condition of the fifth test. The commitments follow from the tests: to publish its language coverage, to document adaptations from pilots, to publish its data practices, to keep core components open, and to work through Kenyan universities, training institutions and employers.

9. Conclusion

To say that digital learning should be local is to make a claim about effectiveness and sustainability, not about identity. The evidence supports it where language, pedagogy and institutions are concerned: learners learn more when they understand the language, teachers teach better when they are supported over time, imported models often fail to take hold, and institutions can only sustain what they can own and adapt. The claim is limited by the thinness of direct evidence on locality as such, and by the risk that locality becomes a slogan. The remedy is to turn it into tests that can be failed, and to publish the results. A provider that does so earns the trust that the word is meant to signal.

Statement on method and interests

This working paper is a narrative review. Sources were identified through web and repository searches in October 2026, giving priority to international agency reports, systematic reviews and randomised or quasi-experimental studies. Bibliographic details were checked against publisher or repository records; for several sources only the abstract or catalogue record was available, and statements here are limited to what those records report. Statements about the content of Kenyan statutes and of continental strategies are summarised at a general level and should be checked against the primary text before being relied upon in legal or policy settings. Drafting used AI assistance (Claude, Anthropic), and the author is responsible for the content. The author is the founder of Instracta eLearning Solutions, whose products and partnerships are discussed. The paper has not been peer reviewed.

References

African Union. (2016). Continental education strategy for Africa 2016–2025 (CESA 16-25). African Union Commission.

African Union. (2020). The digital transformation strategy for Africa (2020–2030). African Union Commission.

Andrews, M., Pritchett, L., & Woolcock, M. (2017). Building state capability: Evidence, analysis, action. Oxford University Press.

Bold, T., Filmer, D., Martin, G., Molina, E., Rockmore, C., Stacy, B., Svensson, J., & Wane, W. (2017). Enrollment without learning: Teacher effort, knowledge, and skill in primary schools in Africa. Journal of Economic Perspectives, 31(4), 185–204.

Digital Public Goods Alliance. (n.d.). Digital Public Goods Standard. https://digitalpublicgoods.net/standard

Hennessy, S., D'Angelo, S., McIntyre, N., Koomar, S., Kreimeia, A., Cao, L., Brugha, M., & Zubairi, A. (2022). Technology use for teacher professional development in low- and middle-income countries: A systematic review. Computers and Education Open, 3, Article 100080. https://doi.org/10.1016/j.caeo.2022.100080

ITWeb Africa. (2022, April 13). Kenya launches ten-year ICT masterplan to capitalise on innovation. https://itweb.africa/article/kenya-launches-ten-year-ict-masterplan-to-capitalise-on-innovation/lwrKxq3KpPoqmg1o

Ladson-Billings, G. (1995). Toward a theory of culturally relevant pedagogy. American Educational Research Journal, 32(3), 465–491.

Ngũgĩ wa Thiong'o. (1986). Decolonising the mind: The politics of language in African literature. James Currey.

Ouane, A., & Glanz, C. (Eds.). (2011). Optimising learning, education and publishing in Africa: The language factor. UNESCO Institute for Lifelong Learning and Association for the Development of Education in Africa.

Piper, B., Simmons Zuilkowski, S., & Mugenda, A. (2014). Improving reading outcomes in Kenya: First-year effects of the PRIMR Initiative. International Journal of Educational Development, 37, 11–21. https://doi.org/10.1016/j.ijedudev.2014.02.006

Piper, B., Zuilkowski, S. S., Dubeck, M., Jepkemei, E., & King, S. J. (2018). Identifying the essential ingredients to literacy and numeracy improvement: Teacher professional development and coaching, student textbooks, and structured teachers' guides. World Development, 106, 324–336.

Pritchett, L., & Woolcock, M. (2004). Solutions when the solution is the problem: Arraying the disarray in development. World Development, 32(2), 191–212.

Pritchett, L., Woolcock, M., & Andrews, M. (2013). Looking like a state: Techniques of persistent failure in state capability for implementation. Journal of Development Studies, 49(1), 1–18.

Republic of Kenya. (2010). The Constitution of Kenya, 2010. National Council for Law Reporting.

Republic of Kenya. (2013). Basic Education Act, No. 14 of 2013. Government Printer.

Republic of Kenya. (2019). Data Protection Act, No. 24 of 2019. Government Printer.

Schweisfurth, M. (2013). Learner-centred education in international perspective: Whose pedagogy for whose development? Routledge.

Tabulawa, R. (2003). International aid agencies, learner-centred pedagogy and political democratisation: A critique. Comparative Education, 39(1), 7–26.

UNESCO. (2016). If you don't understand, how can you learn? Global Education Monitoring Report policy paper. UNESCO.

UNESCO. (2019). Recommendation on Open Educational Resources (OER). UNESCO General Conference, 40th session.

UNESCO. (2023). Global education monitoring report 2023: Technology in education. A tool on whose terms? UNESCO.

How to cite

Ouru, O. M. (2026). Built Where It Is Used: Language, pedagogy, open technology and local partnership as conditions of effective digital learning in Africa. Instracta Research Papers, No. 3. Instracta eLearning Solutions. https://instracta.com/research/local

Approximately 2300 words excluding references. This paper is a working paper and has not been peer reviewed. Comments and corrections are welcome through the Contact page.